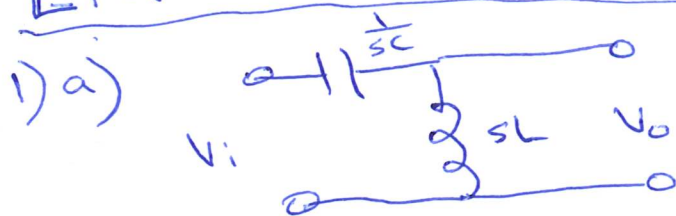


ENMT 301 - Sensors & Signals Tutorial 3 Solutions



$$H(s) = \frac{V_o(s)}{V_i(s)}$$

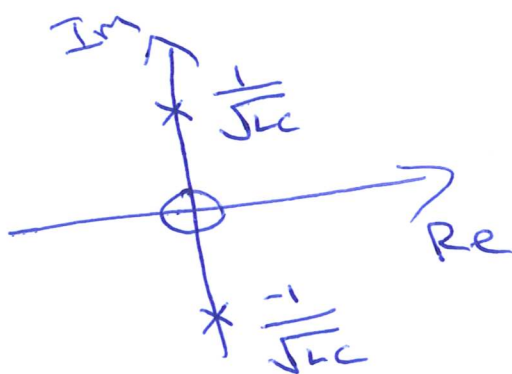
Use voltage divider in s domain

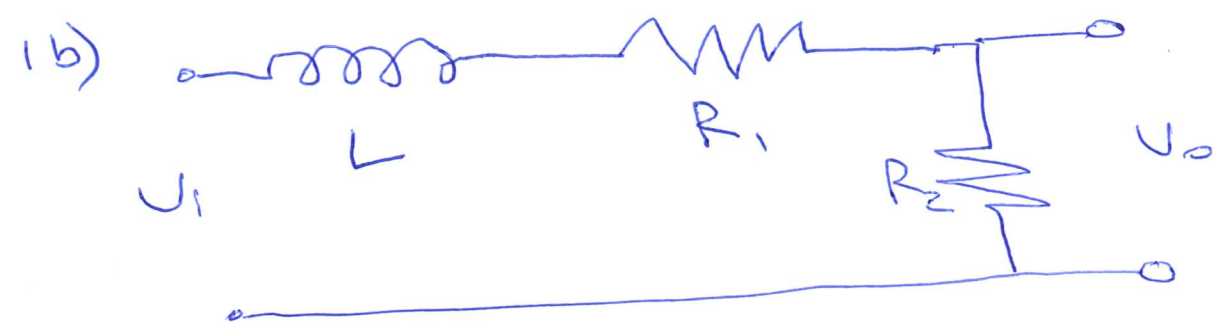
$$V_o(s) = \frac{sL}{sL + \frac{1}{sC}} V_i(s)$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{sL}{sL + \frac{1}{sC}} \quad \downarrow \times \frac{s/L}{s/L}$$

$$H(s) = \frac{s^2}{s^2 + \frac{1}{LC}}$$

Two zeros at $s=0$
Two poles where $s^2 = -\frac{1}{LC}$, $s = \frac{\pm\sqrt{-1}}{\sqrt{LC}} = \pm j \frac{1}{\sqrt{LC}}$





$$V_o(s) = \frac{R_2}{sL + R_1 + R_2} V_i(s)$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{R_2}{sL + R_1 + R_2}$$

$$H(s) = \frac{R_2/L}{s + \left(\frac{R_1 + R_2}{L}\right)}$$

no zeros

one pole at $-\left(\frac{R_1 + R_2}{L}\right)$



2) a) $H(s) = \frac{0.5}{s+0.5}$ Transfer function
Sub. $s = j2\pi f$ to obtain Frequency response

$$H(f) = \frac{0.5}{j2\pi f + 0.5}$$

$$H(f) = \frac{0.5}{j2\pi f + 0.5} \times \frac{0.5 - j2\pi f}{0.5 - j2\pi f} \quad \times \text{complex conjugate of denominator}$$

$$H(f) = \frac{0.5(0.5 - j2\pi f)}{0.5^2 + (2\pi f)^2}$$

$$\operatorname{Re}\{H(f)\} = \frac{0.5 \times 0.5}{0.5^2 + (2\pi f)^2} \quad \operatorname{Im}\{H(f)\} = \frac{-0.5 \times 2\pi f}{0.5^2 + (2\pi f)^2}$$

$$|H(f)| = \sqrt{\operatorname{Re}\{H(f)\}^2 + \operatorname{Im}\{H(f)\}^2}$$

$$|H(f)| = \frac{0.5}{0.5^2 + (2\pi f)^2} \cdot \sqrt{0.5^2 + (2\pi f)^2}$$

$$|H(f)| = \frac{0.5}{\sqrt{0.5^2 + (2\pi f)^2}}$$

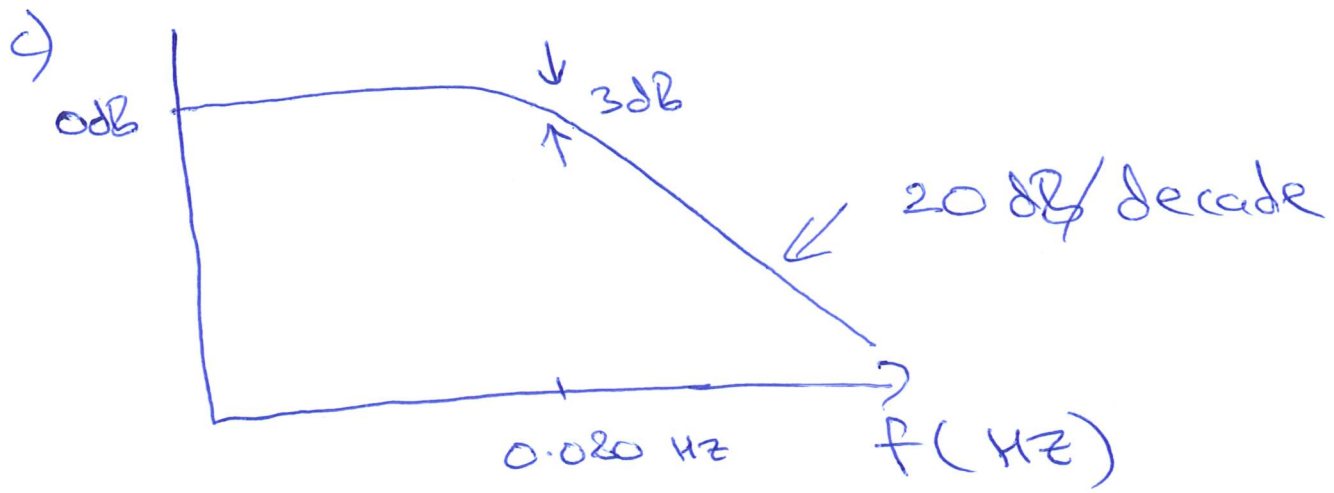
b) $|H(f)| = \frac{1}{\sqrt{2}}$ for 3dB cut-off frequency

$$\frac{0.5}{\sqrt{0.5^2 + (2\pi f)^2}} = \frac{1}{\sqrt{2}}$$

so $(2\pi f)^2 = 0.5^2$

$$2\pi f = 0.5$$

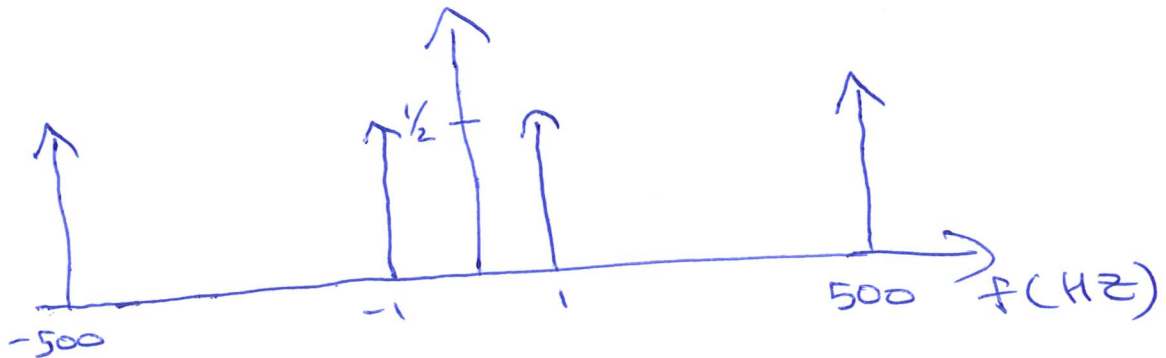
$$f = \frac{0.5}{2\pi} \quad \text{or} \quad \frac{1}{4\pi} \quad \text{or} \quad 0.080 \text{ Hz}$$



d)

$$x(t) = \cos(2\pi t) + \cos(1000\pi t)$$

$$\cos(2\pi f_0 t) \longleftrightarrow \frac{1}{2} \delta(f - f_0) + \frac{1}{2} \delta(f + f_0)$$

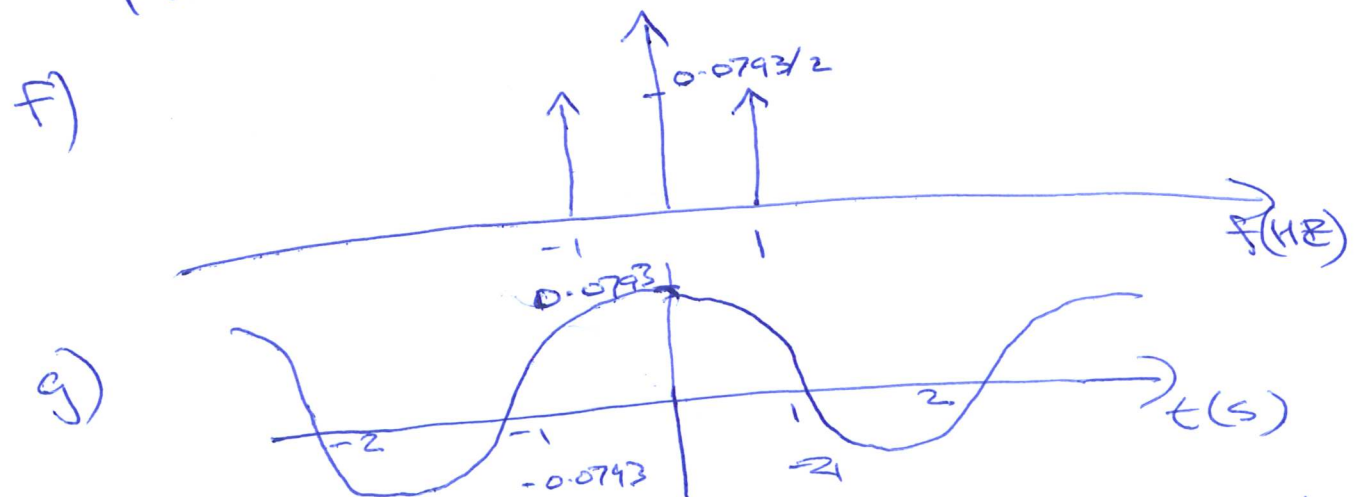


e)

$$|H(1)| = \frac{0.5}{\sqrt{0.5^2 + (2\pi)^2}} = 0.0793$$

$$|H(500)| = \frac{0.5}{\sqrt{0.5^2 + (1000\pi)^2}} = 0.000159$$

$$\frac{|H(1)|}{|H(500)|} = \frac{0.0793}{0.000159} = 498 \text{ or } 54 \text{ dB}$$

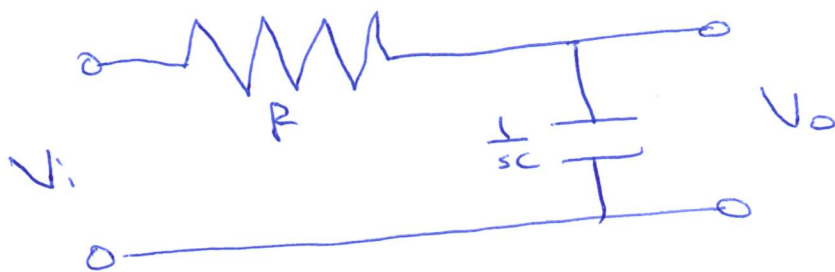


h)

$$|H(0)| = \frac{1}{0.5} = 2$$

The DC gain is greater than zero which cannot be achieved with passive components.

3) a)



$$b) V_o(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} V_i(s)$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{sCR + 1}$$

$$H(s) = \frac{1/Rc}{s + 1/Rc}$$

Transfer function

$$c) H(f) = \frac{1/Rc}{j2\pi f + 1/Rc}$$

Frequency Response

$$H(f) = \frac{1/Rc}{j2\pi f + 1/Rc} \times \frac{1/Rc - j2\pi f}{1/Rc - j2\pi f}$$

$$H(f) = \frac{1/Rc (1/Rc - j2\pi f)}{(1/Rc)^2 + (2\pi f)^2}$$

$$|H(f)| = \frac{1/Rc}{\sqrt{(1/Rc)^2 + (2\pi f)^2}}$$

$$d) |H(f)| = \frac{1}{\sqrt{2}} \text{ at } -3dB \text{ point}$$

$$\therefore 2\pi f_c = \frac{1}{Rc}$$

$$f_c = \frac{1}{2\pi Rc}$$

$$R = \frac{1}{2\pi f_c c}$$

$$R = \frac{1}{2\pi \times 20 \times 10^3 \times 1 \times 10^{-9}}$$

$$R = 7960 \Omega$$

e) Maximum gain of this low-pass filter is at DC (0 Hz)

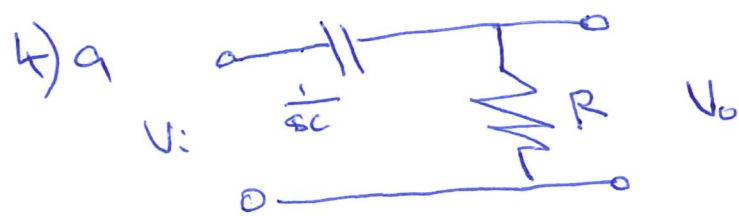
$$|H(\omega)| = \frac{1/R_C}{\sqrt{(1/R_C)^2}} = 1 \quad \text{or } 0 \text{ dB}$$

$$|H(2 \times 10^6)| = \frac{1/(7960 \times 1 \times 10^{-9})}{\sqrt{(1/(7960 \times 1 \times 10^{-9}))^2 + (2 \times \pi \times 2 \times 10^6)^2}}$$

$$|H(2 \times 10^6)| = 0.010$$

$$\text{Attenuation} = \frac{1}{0.010} = 100 \quad \text{or } 40 \text{ dB}$$

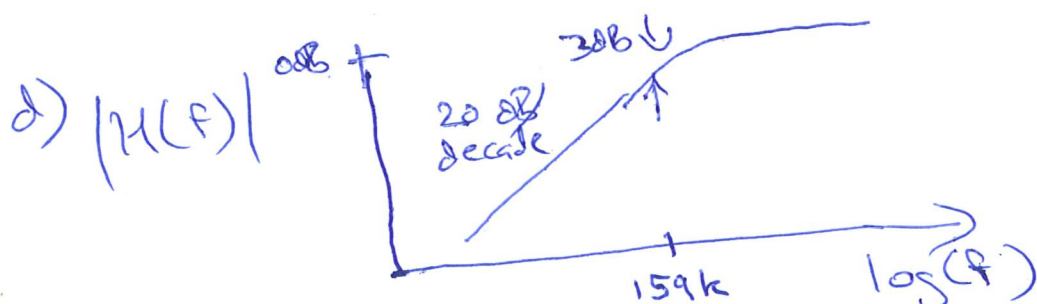
f) Lower the cut-off frequency or increase the order of the filter.



$$b) V_o(s) = \frac{R}{R + \frac{1}{sC}} V_i(s)$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{R}{R + \frac{1}{sC}} = \frac{sCR}{sCR + 1} = \frac{s}{s + \frac{1}{RC}} = \frac{s}{s + 1 \times 10^6}$$

$$c) f_c = \frac{1}{2\pi RC} = \frac{1}{2 \times \pi \times 1 \times 10^3 \times 10^{-6}} = 159 \text{ kHz}$$



$$e) H(f) = \frac{j2\pi f}{j2\pi f + 2\pi f_c} \quad \text{for } f \ll f_c \quad H(f) \approx \frac{jf}{f_c} = \frac{\angle 90^\circ}{\angle 0^\circ} = 90^\circ \text{ or } \pi/2$$